USN | | | | | | | | | | [22TBSCSMA61]

SIXTH SEMESTER B.Tech EXAMINATION, FEBRUARY - 2025
TIME SERIES ANALYSIS AND FORECASTING TECHNIQUES

Time: 3 Hours **Maximum Marks: 100**

Instructions: **(i) Missing data may be suitably assumed**
(ii) Answer all Questions

			Blooms Level	CO	Marks
		PART – A			**5 x2=10**
1.	**a.**	Define time series with a suitable example.	L1	CO1	**02**
	b.	Mention the methods of measuring seasonal variations.	L1	CO2	**02**
	c.	Write the interpretation of regression line with slope being zero.	L2	CO3	**02**
	d.	Define forecasting techniques with suitable examples.	L1	CO4	**02**
	e.	Write the AR and MA polynomial for the given ARMA model $Y_t = 0.7Y_{t-1} - 0.6Y_{t-2} + Z_t - 2.3Z_{t-1} + 0.8Z_{t-2} + Z_{t-3}$.	L2	CO6	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Determine the trend of the following data and estimate the sale for the year 2000, by using semi-average method.	L3	CO1	**05**

Year	1993	1994	1995	1996	1997	1998
Sales (1000's)	20	24	22	30	28	32

OR

	b.	Fit a trend line by using freehand method for the given data and also mention the demerits of this method.	L3	CO1	**05**

Year	1972	1973	1974	1975	1976	1977	1978
No. of accidents	2493	2638	2699	3038	3745	4079	4688

3.	**a.**	Compute the seasonal index for the following data assuming that there is no need to adjust the data for the trend.	L3	CO2	**05**

Quarter	1994	1995	1996	1997	1998
I	78	76	72	74	76
II	66	74	68	70	74
III	84	82	80	84	86
IV	80	78	70	74	82

OR

	b.	The seasonal indices of the sales of garments of a particular type in a certain shop are given below:	L3	CO2	**05**

Quarter	Jan-Mar	Apr-June	July-Sept	Oct-Dec
Seasonal Index	97	85	83	135

If the total sales in the first quarter of a year be worth Rs. 15,000 and sales are expected to rise by 4% in each quarter, determine how

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		much worth of garments of this type be kept in stock by the shop-owner to meet the demand for each of three quarters of the year.			
4.	a.	Find the least square regression line for the following set of data {(-1, 0), (0, 2), (1, 4), (2, 5)} and estimate the value of y when x = 10.	L3	CO3	**05**
		OR			
	b.	Find the regression line of y on x for the following data.	L3	CO3	**05**

x	1	2	3	4
y	2	4	6	8

5.	a.	Discuss the difference between ACF and PACF.	L2	CO4	**05**
		OR			
	b.	Explain the concept of random walk model in time series analysis.	L2	CO4	**05**
6.	a.	What is a white noise process? Discuss its significance.	L2	CO4	**05**
		OR			
	b.	What is a stationary time series? Explain its different types with appropriate examples.	L2	CO5	**05**
7.	a.	Compute the exponential smoothing forecasts for the given time series considering $\alpha = 0.2$ and compute the forecast for the 7th week.	L3	CO6	**05**

Week	1	2	3	4	5	6
Time series value	18	13	16	11	17	14

		OR			
	b.	For the following data, calculate the Mean absolute error by using four-monthly moving average and forecast the value for 8th month.	L3	CO6	**05**

Month	1	2	3	4	5	6	7
Time series value	24	13	20	12	19	23	15

		PART - C		**6 X 10=60**	
8.	a.	A grocery shop owner in a small colony gets 50 bread loaves every morning from a company and sells them to the residents of that colony. If he sells a loaf, he makes a profit of Rs.2/-. If a loaf is not sold on the same day, the shop owner returns it to the company the next morning but he loses Rs. 11- on each loaf returned. In order to maximize his profits, the shop owner decided to study the pattern of the demand for the bread. (The daily demand for bread can be thought of as a random variable.) He collected the data shown in the table below:	L3	CO1	**10**

Day	Jan 1	Jan 8	Jan 15	Jan 22	Jan 29
Sun	45	52	55	56	64
Mon	46	53	59	56	60
Tue	48	45	46	51	57
Wed	47	51	53	58	55
Thu	58	61	60	66	73
Fri	58	62	61	61	70
Sat	51	56	65	64	65

Estimate the value for the sixth week on Sunday.

OR

	b.	The following are the monthly figures of advertising expenditure and sales of a firm. It is generally found that advertising expenditure has its impact on sales generally after 2 months. For this time lag, calculate the coefficient of correlation.	L3	CO1	**10**

Months	Advertising expenditure	Sales	Months	Advertising expenditure	Sales
Jan	50	1200	Jul	140	2400
Feb	60	1500	Aug	160	2600
Mar	70	1600	Sep	170	2800
Apr	90	2000	Oct	190	2800
May	120	2200	Nov	200	3100
Jun	150	2500	Dec	250	3900

9.	**a.**	Explain Link Relative method and also mention its merits and demerits.	L2	CO2	**10**
		OR			
	b.	What are the various methods used for measuring cyclical variations? Explain Reference cycle analysis method of measuring cyclical variations.	L2	CO2	**10**
10.	**a.**	For the following data, estimate the equation that describes the relationship between the age of the truck and its annual repair expense. Also calculate the sum of squared error.	L3	CO3	**10**

Truck No.	Age of the truck in years (X)	Repair expenses during the last years (Y)
101	5	7
102	3	7
103	3	6
104	1	4

		OR			
	b.	Determine the regression equation for the following data that models the relationship between Mathematics scores and happiness levels (scaled from 1 to 5). Further, predict the happiness level for a score of 85, and provide an interpretation of the findings.	L3	CO3	**10**

Maths Score (X)	40	60	70	80	90	100
Happiness (Y)	2	3	4	4	5	5

11.	**a.**	Given the following Sales for seven weeks:	L3	CO4	**10**

Week	1	2	3	4	5	6	7
Sales	39	44	40	45	38	43	39

Forecast the sales for the 8th week using the Simple Moving Average (SMA) method with a window size of 3 weeks. Evaluate the forecast accuracy for years 4, 5, 6 and 7 by calculating MAD (Mean absolute deviation) /MAE, MSE, RMSE, MAPE and also analyze whether the forecasting method tends to overestimate or underestimate the actual sales.

		OR			
	b.	What do you mean by Depersonalization in Time series Analysis? Explain the methods and applications of depersonalization with suitable examples.	L2	CO4	**10**
12.	**a.**	Discuss the conditions for stationarity and non-stationarity for a time series data and interpret whether $X_t = Z_t Z_{t-1}$ is stationary or not, assuming $\{Z_t\}$ as a sequence of independent, identically distributed (i.i.d.) random variables with $E[Z_t] = 0$ and $Var[Z_t] = \sigma^2$.	L3	CO5	**10**
		OR			
	b.	Let $\{Z_t\}$ be a sequence of independent normal random variables with mean 0 and variance σ^2, let a and b be constants. Check which of the following processes are stationary and also specify their mean, variance and autocovariance. **i.** $X_t = a\cos(bt) + Z_t$ **ii.** $X_t = a + bZ_t$	L3	CO5	**10**
13.	**a.**	**i.** For the given ARMA model: $X_t = 0.75X_{t-1} - 0.56X_{t-2} + Z_t - 1.3Z_{t-1} + 0.3Z_{t-2}$ where, X_i are AR terms and Z_i are MA terms. **ii.** Identify the order of the model. Check the stationarity and invertibility conditions for this model.	L3	CO6	**10**
		OR			
	b.	**i.** Define ARIMA Model and write the general polynomial of AR and MA process obtained from ARIMA. **ii.** Identify the order of given ARIMA model $X_t = 1.5X_{t-1} - 0.5X_{t-2} + Z_t - Z_{t-1} + 0.25Z_{t-2}$ where, X_i are AR terms and Z_i are MA terms.	L3	CO6	**10**